虚拟机python环境

因为conda装的python环境和官网的有不同，所以以官网方式重新装一下python，以3.7为例

3.7，3.8都行，neo4j都可以跑

```
#检测可更新
sudo apt update
#用于添加ppa源的小工具，ubuntu server默认没装。
sudo apt install software-properties-common

#接下来，将Deadsnakes PPA添加到您的来源列表中：（不做这步也可以，但会安装较老的版本）
#有确认提示按回车继续
sudo add-apt-repository ppa:deadsnakes/ppa

#搜索是否有需要的版本
sudo apt search python3.7
#安装
sudo apt install python3.7

python3.7 -V

#安装curl
sudo apt install curl

#安装pip
# 下载最新的官方 pip
sudo curl https://bootstrap.pypa.io/get-pip.py -o get-pip.py
# 安装 python-distutils
sudo apt install python3-distutils 或 sudo apt install python3.9-distutils(高版本的python3)
# 直接获取 pip3.7
sudo python3.7 get-pip.py
# 安装 launchpadlib
pip3.7 install launchpadlib

#测试是否安装成功
pip3.7 --version

#如果有遇到 C 实现的 pip 安装不了，则需要安装。
sudo apt-get install python3.7-dev

#查询路径
which python3.7
which pip3.7
```

之后安装neo4j，该关代理关代理

```
pip install neo4j
```

查看neo4j安装路径，将neo4j包移动到项目下，不然py文件无法链接neo4j

```
pip show neo4j
cd /home/zyzhu/.local/lib/python3.8/site-packages
sudo cp -r neo4j /tmp/IROS/
```

如果/usr/include/python3.10下没有Python.h，需要安装环境

```
sudo apt-get install python3-dev
```

cmakelist.txt里面include_directories(/usr/include/python3.10)

neo4j配置

打开conf目录下的配置文件neo4j.conf

添加两行，不然虚拟机里面无法连接neo4j

```
dbms.connector.http.listen_address=0.0.0.0:7474
dbms.connector.bolt.listen_address=0.0.0.0:7687
```

Clion配置

配置cmakelist，路径都是虚拟机里python路径

```
cmake_minimum_required(VERSION 3.0)
project(IROS)

set(CMAKE_CXX_STANDARD 14)
include_directories(/usr/include/python3.8)
link_directories(/usr/lib/python3.8/config-3.8-x86_64-linux-gnu)

add_executable(IROS main.cpp)
target_link_libraries(IROS libpython3.8.so)
```

main函数

```cpp
#include <iostream>
//#include <python3.8/Python.h>
#include <Python.h>

using namespace std;
int main() {
    cout << "Hello, World!" <<endl;
    cout << "1"<<endl;
//    Py_Initialize();  //初始化python环境
//    PyRun_SimpleString("print('hello world!')");  //使用print语句
//    Py_Finalize();  //结束调用
    Py_Initialize();//使用python之前，要调用Py_Initialize();这个函数进行初始化

    if (!Py_IsInitialized())
    {
        printf("初始化失败！");
        return 0;
    }

    PyObject* pModule = NULL;//声明变量
    PyObject* pFunc1 = NULL;// 声明变量
    PyObject* pFunc2 = NULL;// 声明变量
    PyObject* pFunc3 = NULL;// 声明变量
    PyRun_SimpleString("import sys");
    PyRun_SimpleString("sys.path.append('/tmp/IROS/')");
    pModule = PyImport_ImportModule("ttest");//这里是要调用的文件名
    if (pModule == NULL)
    {
        cout << "没找到" << endl;
    }

    pFunc3 = PyObject_GetAttrString(pModule, "getMessage");//这里是要调用的函数名
    PyObject* pArgs = PyTuple_New(1);  //创建元组对象,声明参数个数为2
    PyTuple_SetItem(pArgs, 0, Py_BuildValue("s", "RGBCap1"));  //第0个参数，int类型，值a
    PyObject* args2 = Py_BuildValue("s", "RGBCap1");//给python函数参数赋值

    PyObject* pRet = PyObject_CallObject(pFunc3, pArgs);//调用函数
    char *res = NULL;
//    int v = -1;
    if(pRet == NULL){
        cout << "null" << endl;//输出结果
    } else{
        cout << "不null" << endl;//输出结果
    }
//    PyArg_Parse(pRet, "i", &v);
    PyArg_Parse(pRet, "s", &res);//转换返回类型

    cout << "res:" << res << endl;//输出结果
```

```
    Py_Finalize(); // 与初始化对应
//    pause();
    return 0;
}
```

需要注意的是，接受字符串类型的变量只能下面这样：

```
char *p=NULL;
PyArg_Parse(pRet, “s”, &p);
```

需要把ttest.py放到main.cpp同级目录
ttest.py

```
from neo4j import GraphDatabase

driver = GraphDatabase.driver("neo4j://10.71.243.200:7687",
                              auth=("neo4j", "zhangyu123"))
stringVal = '你好'
def add_friend(tx, name, friend_name):
    tx.run("MERGE (a:Person {name: $name}) "
           "MERGE (a)-[:KNOWS]->(friend:Person {name: $friend_name})",
           name=name, friend_name=friend_name)

def print_friends(tx, name):
    query = ("MATCH (a:Person)-[:KNOWS]->(friend) WHERE a.name = $name "
             "RETURN friend.name ORDER BY friend.name")
    for record in tx.run(query, name=name):
        print(record["friend.name"])

def get(tx,message):
    query = ("match (n:ns0__ImageCapability) where n.ns0__name = $message "
             "return n")
    # print(tx.run(query, message=message))
    for record in tx.run(query, message=message):
        # print(record)
        stringVal = record["n"]
        print(record["n"])
        # return record["n"]

def getMessage(message):
    with driver.session(database="neo4j") as session:
        # session.execute_write(add_friend, "Arthur", "Guinevere")
        # session.execute_write(add_friend, "Arthur", "Lancelot")
        # session.execute_write(add_friend, "Arthur", "Merlin")
        # session.execute_read(print_friends, "Arthur")
```

```
            session.execute_read(get, message)
            return stringVal
            # session.execute_read(get,"RGBCap1")

    driver.close()
```

没有Python.h头文件

_sudo a_pt-get install python3-dev

pModule == NULL，评估：没找到python文件

安装或者更新numpy

```
sudo apt-get install python3-dev
pip config set global.index-url https://pypi.tuna.tsinghua.edu.cn/simple
pip install numpy
```

wheatfox

```
python -m pip install --upgrade --no-cache-dir --use-deprecated=legacy-resolver
<your_package>
pip install -U pip # whis worked
```